Action, Trajectories and Gaps: A Research Programme

navstokgap research working notes

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Abstract

We organize the study of mechanical action scales around exact trajectories, fluctuation spectra and operational resolution. A sequence from constant force through oscillator and central-potential models leads to two research targets: quantum reconstruction and a mechanism for gap formation. Each work package has a mathematical object, deliverable and acceptance gate. Relativistic and PDE/field-theory comparisons track how estimates behave under changes of scale. The maintained source is `research/PROGRAMME.md`; task state and handoffs support continuation across research sessions.

1 Research programme: action, trajectories and gaps

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1.1 Aim

Explain the relationship between mechanical trajectories, an action scale and gap formation. We pursue two connected outcomes:

1. A reconstruction of quantum mechanics from explicit physical premises, with a derivation of the role and necessity of an action parameter.
2. A rigorous toy model that identifies which Lagrangians, state spaces and boundary conditions produce a positive lower bound.

Newton's shrinking-area construction provides the initial geometric model. The work proceeds through exact mechanical examples, operational definitions, operator spectra and comparisons of limiting procedures.

1.2 Starting results

For $V = -Fy$, duration $T > 0$, perpendicular launch speed $v_0 > 0$ and mass $m > 0$, the chord–curve action difference is

$$\Delta S = \frac{F^2T^3}{24m} = \frac{F}{2v_0}\mathcal{A}_{\mathrm{lens}} = \frac{T\delta E}{12}, \qquad \delta E = \Delta K = -\Delta V.$$

The variation $\eta(t) = at(T-t)$ gives $\Delta S = ma^2T^3/6$. Thus positive action differences in this fixed-endpoint class have infimum zero. The chord is an off-shell comparison path; the parabola solves the constant-force equation. This identifies admissibility of neighbouring paths as a substantive choice for the proposed gap mechanism.

A second result concerns quantum measurement. For two known, equally likely two-arm pure states whose relative phase is $\Delta S/\hbar$, N independent copies and arbitrary joint binary measurements give the first-lobe threshold

$$d_{N,p} = 2\hbar \arccos\Big([4p(1-p)]^{1/(2N)}\Big), \qquad |\Delta S| \le \pi\hbar, \quad \tfrac{1}{2} < p \le 1.$$

Success probability at least p is achievable exactly when $|\Delta S| \geq d_{N,p}$. For fixed $\frac{1}{2} < p < 1$, this threshold scales as $N^{-1/2}$; perfect discrimination has threshold $\pi\hbar$ for every finite N. The calculation connects an action difference to a resource-dependent resolution.

The technical paper supplies the proofs, the Jacobi-operator calculation and the normalized free-kernel limit. The ledger records their evidence and review status.

1.3 Questions and mathematical objects

Target	Question	Starting point
Q0: classical existence	Which force regularity, initial data and collision conventions give unique global trajectories?	Constant force; next, central potentials
Q1: action-value gap	Is the infimum of positive matched-endpoint action differences positive on the chosen path class?	Infimum zero for the constant-force family
Q2: action scale	Which physical premises force a universal parameter with action units to be positive?	Reconstruction axiom comparison
Q3: operational bound	How do protocol, resources and error target determine action resolution?	Exact two-arm finite-copy threshold
Q4: spectral gap	Which self-adjoint operator has a separated ground sector, and how does its gap depend on parameters?	Dirichlet Hessian and quantum Hamiltonian
Q5: transfer across limits	Which estimates survive continuum and infinite-volume limits and interactions?	Companion PDE/field-theory comparisons

Use $h = 2\pi\hbar$ for Planck's constants, η for trajectory variations, J for a fluctuation operator and χ for a proposed additional field. Use matched endpoints when comparing actions under additions of $dG(q,t)/dt$.

1.4 Model sequence

1. **Constant force:** derive the area–action relation and study small variations.
2. **Harmonic oscillator:** compare the duration-dependent Hessian eigenvalues with the quantum Hamiltonian spacing. Track normalization and conjugate times.
3. **Free line, circle and box:** isolate confinement and boundary conditions, and calculate gap closure as the spatial domain grows.
4. **Central potentials and Kepler:** establish IVP assumptions, collision continuation and action-angle variables; then compare quantum spectra.
5. **Interacting and multiwell models:** identify a gap-producing mechanism and its dependence on coupling, tunnelling and limiting parameters.

1.5 Work packages

Package	Deliverable / acceptance gate	Dependencies
WP0: foundations	Restart instructions, ledger, tasks, source rules and LaTeX/PDF build	Complete
WP1: Newton	Edition/folio concordance, six-scholium corpus and dated evidence for the publication-history question	Source access
WP2: classical mechanics	Proofs for constant force, regular central forces and specified Kepler collision cases	Path class and topology
WP3: fluctuations and kernels	Jacobi operator/domain, conjugate times and normalized distributional limits	WP2
WP4: operational reconstruction	Measurement bounds and a dependency map of candidate quantum axioms	WP3; bibliography B01
WP5: gap laboratory	Operator domains, explicit bounds and parameter-dependent gap closure	Relevant WP2–4 results
WP6: companion limits	Relativistic, parabolic and quantum-field comparisons with a specified transferable estimate	WP5; Millennium definitions
WP7: papers and review	Readable proofs, checked citations, reproduced results and selected formal certificates	Accepted package result

Historical reading and mathematical calculations have separate task tracks. Delegate bounded tasks to Sol or Luna sequentially, with the coordinator waiting for each worker and reviewing its handoff before continuing. The finite-dimensional examples establish the definitions and estimates for the later field-theory comparison. A package finishes with a theorem, counterexample, conditional result or precisely stated open obligation.

1.6 Reconstruction branches

Branch A: consequences of quantum premises. Start with Hilbert states, Born probabilities and the phase $e^{iS/\hbar}$ for a specified $\hbar > 0$. Derive operational bounds, spectra and their dependence on model parameters.

Branch B: origin of quantum structure. Choose operational premises about composition, continuity, reversible transformations and distinguishability. Construct models satisfying subsets of these premises and identify the step that selects quantum structure. Then establish how action enters the dynamics and what forces its scale to be positive. Numerical calibration is a further empirical task.

Additional-field proposal. Specify the field space, coupling and measure for $S[q, \chi]$, then calculate the effective action after eliminating χ. The idea register sets out the candidate interpretations and their immediate tests.

1.7 Reading priorities

Start Q01 with Hardy's five axioms and Chiribella–D'Ariano–Perinotti's informational derivation. Use Masanes–Müller as a third comparison. For each, trace how its premises constrain state spaces, transformations and composition.

Branch A draws on Feynman's path-phase formulation, Wootters's finite-sample statistical distance and the quantum Chernoff asymptotic error bound. The B01 batch records verified metadata and selected reading coverage. Full reconstruction-proof reading belongs to Q01.

1.8 $1/c$, Navier–Stokes and Yang–Mills

Track $\hbar$, c^{-1}, viscosity ν, time and volume independently. The free relativistic particle retains the small-amplitude action family within a strict speed margin. The nonrelativistic quantum model retains $\hbar > 0$ at $c^{-1} = 0$. These examples separate the roles of the two parameters.

A bridge must specify its source and target spaces, operators, physical or auxiliary time, units, preserved estimate and limits. The useful comparison is how coercivity and control of fluctuations survive changes of scale. For Navier–Stokes, the target is evolution regularity under the official hypotheses. For Yang–Mills, it includes the physical Hilbert space and a vacuum spectral gap through the continuum and infinite-volume limits.

1.9 Scope and publication

The programme's reconstruction and gap-formation questions are open research targets. The present results concern the explicitly defined toy models; the Millennium problems supply comparison targets. Novelty and publication readiness will be assessed through literature comparison and specialist review.

Maintain human-readable proofs alongside calculations and formal artifacts. Record changes of model or target with their reasons. Submission, public posting, author correspondence and paid services require user direction.